



MMA Companion

Computer Science NEA

Thomas Jubb

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6.1 Testing Strategy

6.11 Approach

Functional testing

A lot of the testing I have done was developmental during the implementation of the features themselves. This could mean that some features may have been broken during development due to changes in the main class, which could be due to things like severed attributes between methods, methods no longer being linked to the UI due to an object in the UI being named etc. For non-complex calculations like the statistics on pie charts, I'll calculate the result myself and then compare it to what the program shows. For the more complex calculations like the ELO calculations, I'll use the help of an LLM like ChatGPT or Copilot to calculate what the updates should look like.

Validation testing

I am going to make sure that all data is validated like the data dictionary in the design/PPROD documents suggest. I'm going to do this by testing the data inputs the system offers and inputting typical data that should be considered valid to see how the system normally treats data, extreme data that is on the line between valid and invalid to see how the system handles the transition between valid and invalid data and erroneous data to see how robust the system is and how it treats invalid data.

Data handling

For the testing I am going to make typical data tables with dummy data. This data will all be typical.

6.2 Testing

6.21 Functional Testing plan

Test ID	Description	Type Of Test	Data Used	Expected Outcome
F1a	Sign up and duplicate check	Typical	New username and password	Account is created successfully.
F1b	Sign up and duplicate check	Erroneous	Username that already exists	System rejects signup. Error is shown.
F1c	Sign up validation	Erroneous	Blank username	System rejects signup. Error is shown.
F1d	Sign up validation	Erroneous	Blank password	System rejects signup. Error is shown.
F2a	Login	Typical	Correct username and password	User logs in successfully. Redirected to leaderboard.
F2b	Login	Erroneous	Correct username and wrong password	Login denied and error message shown
F2c	Login	Erroneous	Username doesn't exist	Login denied and error message shown
F3a	First use prompts survey	Typical	New account first login	System redirects to survey
F3b	Survey submission stores results	Typical	Survey stars selected	Values saved to user record and user redirected to leaderboard.
F3c	Redo survey	Typical	Existing user clicks redo survey button in leaderboard	Survey opens then on submit it updates existing survey record (not duplicate insert).
F4a	Favourite fighter selection in survey	Typical	Choose favourite fighter from dropdown box in survey	Dropdown populates and favourite fighter is saved to profile.
F5a	Navigation between pages	Typical	Click navigation buttons	Correct pages open and all others hide.
F6a	Admin views as user	Typical	Admin clicks View as User	Admin sees leaderboard or user pages.
F6b	Return to admin from admin icon	Typical	Modify fighters' button on leaderboard	Admin can return to modify fighter list and admin pages.
F6c	Return to admin button visible for admins	Typical	Admin navigates to leaderboard	Admin sees return to modify fighters button visible
F6d	Return to admin button not visible for users	Typical	User navigates to leaderboard	Users can't see return to modify fighters' button
F7a	Leaderboard displays ranked fighters	Typical	Database shown in prior database handling section	Fighters shown sorted by rating
F7b	Leaderboard sort (quantitative)	Typical	Click quantitative columns	Sort toggles ascending/descending on second click.
F9a	Levenstein name search	Typical	Search "Jon jons" for "Jon Jones"	Returns sorted by closest matches.
F9b	Name search	Erroneous	Blank search	Search denied
F10a	Open fighter profile from leaderboard	Typical	Click/double click fighter's row	Profile loads for the correct fighter.

F11a	Fighter profile shows photo	Typical	Fighter with working image URL	Image loads and displays on profile.
F11b	Fighter profile image loading rework	Boundary	Image URL that takes a while to load	UI stays responsive (threaded image loading).
F12a	Fighter profile shows fight history	Typical	Fighter with past fights	Past fights displays correctly.
F12a	Show fight dates	Typical	Fighter with fights linked to events	Fight table includes event date values.
F13a	Fighter profile shows record and streak	Typical	Fighter with multiple fights	Correct streak displayed
F14a	Fighter profile stats breakdown	Typical	Fighter with mixed outcomes	Method wins/losses displayed
F15a	Belts shown on profile	Typical	Fighter who holds/held belts	Belt list appears correctly.
F16a	Pie chart displays breakdown	Typical	Fighter with a record	Pie chart is displayed correctly for wins/losses and method breakdown.
F16b	Pie chart simplification	Boundary	Fighter has no wins of a certain category	Missing method wins/losses don't appear
F17a	Notable wins displayed	Typical	Fighter has wins vs top 10 ranked in same weight class	Notable wins list displayed
F18a	Height and reach shown	Typical	Fighter with height/reach values	Height/reach appear on profile.
F19a	Odds calculator fighter dropdowns	Typical	Open odds page	Dropdown boxes are populated by weight class or full list if no fighter is selected yet
F19b	Odds dropdown prevents same fighter	Typical	Pick fighter A	Error message shown
F20a	Odds calculation output	Typical	Two fighters of the same weight class	Shows winning odds as percentages
F21a	Entertainment score appears under odds	Typical	User with survey data	Entertainment score is displayed after odds calculation.
F21b	Favourite fighter affects entertainment score	Typical	Favourite fighter is one of selected fighters	Entertainment score increases if the fighter is the user's favourite
F23a	Upvote fighter approval	Typical	User presses positive vote button	Vote stored and approval rating updates.
F23b	Downvote fighter approval	Typical	User presses negative vote button	Vote stored and approval rating updates.
F23c	Prevent duplicate vote	Boundary	User presses positive vote button twice	Second click does not double count, so rating is unchanged on 2nd click.
F23d	Switch vote direction	Typical	User presses positive vote button then negative vote	System re-updates vote
F24a	Admin starts on modify fighter list	Typical	Admin login	Admin redirected to edit fighter list page.

F25a	Add fighter	Typical	Click add fighter button in modify fighter	Blank fighter inserted and it appears in the list.
F25b	Delete fighter	Typical	Select fighter then delete button	Fighter removed from the database and table refreshes.
F25c	No selection on delete fighter	Erroneous	Press delete button with nothing selected	No deletion occurs
F26a	Admin search fighter	Typical	Misspelled name in modify fighter search	Closest matches returned (Levenstein search).
F27a	Open fighters edit page	Typical	Double click fighter row	Fighters edit page opens for the correct fighter.
F28a	Tooltip for fighter IDs in fights table	Typical	Hover over ID cell	Tooltip shows fighter name for that ID.
F29a	Add fight from fighter edit page	Typical	Click add fight button on fighter edit page	Blank fight inserted linked to current fighter.
F29b	Delete fight	Typical	Select fight row then delete	Fight removed safely and the table refreshes.
F30a	Edit fight opponent ID	Typical	Enter valid opponent ID	Opponent row links correctly
F30b	Edit fight opponent ID	Erroneous	Enter opponent ID that doesn't exist	Update rejected and error is shown.
F30c	Edit fight opponent ID	Erroneous	Set opponent ID to be equal to current fighter	Update rejected and error is shown.
F31a	Edit fight result	Typical	Set result to W/L/NC	Update accepted
F31b	Edit fight result formatting	Erroneous	Result in fighter table set to WIN	Update rejected and error is shown
F32a	Submit edits on editable tables	Typical	Change multiple fields then submit	Updates pushed to database and remained after reloading.
F33a	Duplicate detection on submit	Erroneous	Create duplicate fighter/event/fight	System stops submit and shows duplicate to fix.
F33b	Format validation on submit	Erroneous	Invalid method/time/title formats	System blocks submit and lists formatting errors.
F33c	Presence/type checks on submit	Erroneous	Missing required field	System blocks submit and displays missing required fields
F34a	Statistical anomalies check for 0-second fight	Erroneous	Fight duration less than or equal to 1 second	System flags as anomaly and shows errors
F34b	Statistical anomaly for impossible number of yearly fights	Erroneous	Fighter has more than 12 fights in a year	System flags as anomaly and shows errors
F35a	Update ratings	Typical	Database as shown in database handling with a fight added to change ratings	Ratings are recalculated successfully and leaderboard updates to new order.
F35b	Performance check for update	Boundary	Database as shown in database handling with a fight added to change ratings	Update completes within acceptable time.

F36a	Open belts and events options page	Typical	Admin clicks edit belts/events	Page opens and data populated to belts and events table.
F36b	Add belt	Typical	Click add belts button	New blank belt row appears in belts table.
F36c	Delete belt	Typical	Select belt then click delete belts button	Belt deleted and table refreshes
F36d	Delete belt triggers RETIRED title status	Typical	Delete belt for a division	Fights in that division for their former title get title status set to RETIRED.
F37a	Edit belt holder using fighter ID	Typical	Type fighter ID into holder	Holder updates to that fighter.
F37b	Edit belt holder using non-existent fighter ID	Erroneous	Type non existent fighter ID into holder	System shows error
F38a	Add event	Typical	Click add event button	New event inserted and it appears in events table.
F38b	Delete event	Typical	Select event then delete	Event removed and table refreshes.
F39a	Auto-assign belts	Boundary	All belts already assigned	Message shows that no beltless divisions have been found
F39b	Auto-assign belts	Typical	At least one beltless division and a title fight for that division exists	System assigns belt to most recent winner of title fight. Reassignment is displayed

6.22 Functional Testing Results

*For all of the the testing each test is shown in the Testing Videos folder adjacent to this document. Each test is titled by its reference.

Ref	Result and comments
F1a	Pass
F1b	Pass
F1c	Fail because blank validation not present
F1d	Fail because blank validation not present
F2a	Pass
F2b	Pass
F2c	Pass
F3a	Pass
F3b	Pass
F3c	Pass
F4a	Pass
F5a	Pass
F6a	Pass
F6b	Pass
F6c	Pass
F6d	Pass
F7a	Fail because SQL query doesn't sort by rating ,ELO isn't displayed as well
F7b	Pass
F9a	Pass
F9b	Pass
F10a	Pass
F11a	Pass
F11b	Pass
F12a	Pass
F12a	Pass
F12b	Pass
F13a	Pass
F14a	Pass
F14b	Pass
F15a	Pass

Thomas Jubb
3003

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F16a	Pass
F16b	Pass
F17a	Pass
F18a	Pass
F19a	Pass
F19b	Pass
F20a	Pass
F21a	Pass
F21b	Fail as entertainment algorithm tries to divide by zero
F23a	Pass
F23b	Pass
F23c	Pass
F23d	Pass
F24a	Pass
F25a	Pass
F25b	Pass
F25c	Pass
F26a	Pass
F27a	Pass
F28a	Pass
F29a	Pass
F29b	Pass
F30a	Pass
F30b	Pass
F30c	Pass
F31a	Partial pass- works but NC needs to be its own method
F31b	Pass
F32a	Pass
F33a	Partial, duplicates can still happen on fighters and adding also still creates duplicates
F33b	Pass
F33c	Pass

Thomas Jubb
3003

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F34a	Pass
F34b	Fail because not implemented yet. Also, should add more event-based validation.
F35a	Pass
F35b	Pass
F36a	Pass
F36b	Pass
F36c	Pass
F36d	Fail because of bad UPDATE query
F37a	Pass
F37b	Pass
F38a	Pass
F38b	Pass
F39a	Pass
F39b	Pass

Thomas Jubb
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6.22 Validation Testing

Ref	Field	Datatype	Validation	Typical Data	Erroneous Data	Boundary Data 1	Boundary Data 2
Belts							
V1	Weight	Varchar	Length check 60 chars	Welterweight	61 characters	60 characters	1 character
V2	FighterID	Int	Lookup	2	35999033	N/A	N/A
Event							
V3	Name	Varchar	Length check 80 chars	UFC 300	81 characters	80 characters	1 character
V4	Location	Varchar	Length check 100 chars	Madison Square Garden	101 characters	100 characters	1 character
V5	Date	Datetime	Format/range , must be valid and can't be in future	11/03/2024	4th Jan	29/02/2024	(tomorrow's date)
Users							
V6	Username	Varchar	Length	bob123	<blank>	1 char	200 chars
Fights							
V7	EventID	Int	Lookup/type	42	ABC	N/A	N/A
V8	Method	Varchar	Lookup/domain – must be a valid mma result	KO	TKO	SUB	DEC
V9	EndRound	Int	Range/length – must be 1–5	3	7	1	5
V10	EndTime	Varchar	Format/range	2:32	7:12	0:02	5:00
V11	Title	Varchar	Length check	Welterweight	61 chars	60 chars	<blank>
FighterFights							
V12	FighterID	Int	Lookup check	45	999999	N/A	N/A
V13	Result	Varchar	Lookup/domain must be either W L or NC	W	WIN	L	NC
V14	FighterID	Int	Presence check	Existing fighter ID	<blank>	N/A	N/A
Fighters							
V15	Name	Varchar	Length check	Khalil Rountree	61 chars	60 chars	1 char
V16	Birthdate	Varchar(date)	Format/range	11/04/1993	80 years ago	18 years ago	60 years ago
V17	Gym	Varchar	Length check	American Kickboxing Academy	81 chars	80 chars	<blank>

V18	ImageURL	Varchar	Length/format	https://...	(256 chars)	(255 chars) (valid)	<blank>
V19	Height	Float	Format / range	PROCESSED BY SYSTEM 6'00"	6'100"	8'10"	4'0"
V20	Reach	Float	Format / range	PROECESSSED BY SYSTEM 5'10" input	6'20"	8'10"	4'0"

6.22 Validation Testing Results

Ref	Result and comments
V1	Partial - Normal data works but upper boundary generates truncation error and so does the erroneous. Putting in no characters gives an SQL error
V2	Pass
V3	Fail – Erroneous data is submitted and putting in none gives a python error.
V4	Fail – Erroneous data is submitted
V5	Fail – Every entry is format rejected
V6	Fail – blank allowed. SQL error shows on long range but it still goes to survey.
V7	Partial – Rejects input not allowed under type rule but doesn't do a lookup check on events
V8	Pass
V9	Fail – No range validation
V10	Pass
V11	Pass
V12	Pass
V13	Pass
V14	Pass
V15	Partial- Rejected the length check triggering data but also allowed line breaks and blank names
V16	Pass
V17	Pass
V18	Partial – Does length rejections but range is too big
V19	Fail – No range validation lower boundary
V20	Fail – No range validation lower boundary

Thomas Jubb
3003

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Centre: 65131

6.4 Summary and refinements

6.41 Summary and comments

Overall, my functional testing went quite smoothly. Not too many errors were generated, and you can do the core user and admin process without getting errors. The errors that I did get will be patched for the final submitted build. For my validation testing it was quite rocky but identifying the problems with my validation will contribute to the overall robustness of my final build

6.42 Refinements

Issue	Suggested Refinement
No blank catching in login section	Add catch to stop blank usernames and passwords
Leaderboard not sorted by rating	Display ELO and sort by it
Entertainment algorithm tries to divide by zero	Shift the values in the entertainment algorithm to stop it being able to divide by 0
Fighters can have 12 fights on the same day	Reinforce the 12-fight validation and add a week gap requirement between fights
Retired titles are not set to retired	Fix update query
SQL errors that the admin may not be able to figure out are displayed	Create a SQL error handler or add internal systems for database restrictions
Events is not validated on input	Add user input validation to events and add database constraints as well. This should be things like length validation for location and name. It should also have stricter range validation for the date.
All Event dates are rejected	Strip down the entry to avoid accidental entries, split it and rejoin it in the correct format to submit it to the database
Usernames that raise a database error can still get through to the survey	Add login side validation and make the login screen not go through on an SQL error
No events lookup check on fight entry on profile	Add lookup check and database constraints
No range validation	Add 1-5 range to rounds (In database as well)
Finish time formatted badly (fights can't last for hours)	Process end times in fights as mm:ss or more ideally m:ss
Line breaks are allowed on things like names and gym names messing up the UI	Don't allow line breaks to be inputted
Image size length validation range isn't as specified	Adjust image url length constraints to be 256 chars
Height has no lower boundary validation	Add lower boundary validation for height